WHAT ARE NEURAL NETWORKS?

Neural Networks

BUZZWORDS

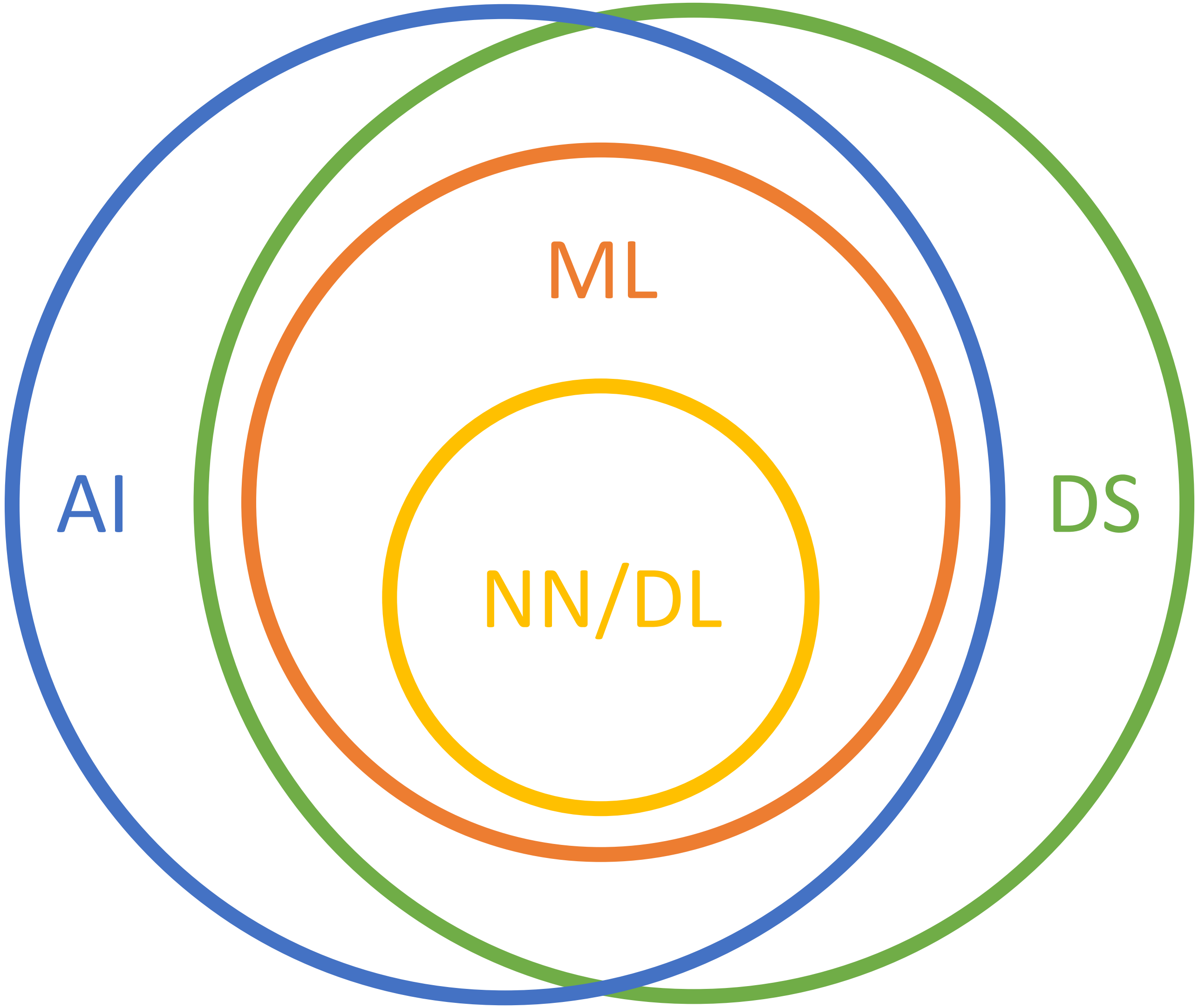
Neural Networks ≠ Data Science

Neural Networks ≠ Machine Learning

Neural Networks ≠ Artificial Intelligence

Neural Networks = Deep Learning

BUZZWORDS



WHAT ARE NEURAL NETWORKS?

Artificial neural networks are machine learning models that are inspired by the biological brain.

WHY NEURAL NETWORKS?

Goal:

Use machines to automate tasks

Challenge:

Solving tasks for which the process of solving is difficult to describe (e.g. image or speech recognition)

WHY NEURAL NETWORKS?

EXPERT SYSTEMS

Definition:

A system which uses domain-specific knowledge to solve a task in a way similar to a human expert.

1. Devise a set of rules that describe a task environment
2. Use reasoning engine to answer questions

EXPERT SYSTEMS

Fred is a human.

Things with electronical components are machines.

Fred shaves with an electric razor.

An electric razor is an electrical component.

Fred is a machine (while shaving).

EXPERT SYSTEMS

- Cannot handle noise
- It is difficult to devise accurate and complete rules
- Assigning things symbols add bias to our interpretation

TRADITIONAL MACHINE LEARNING

Definition:

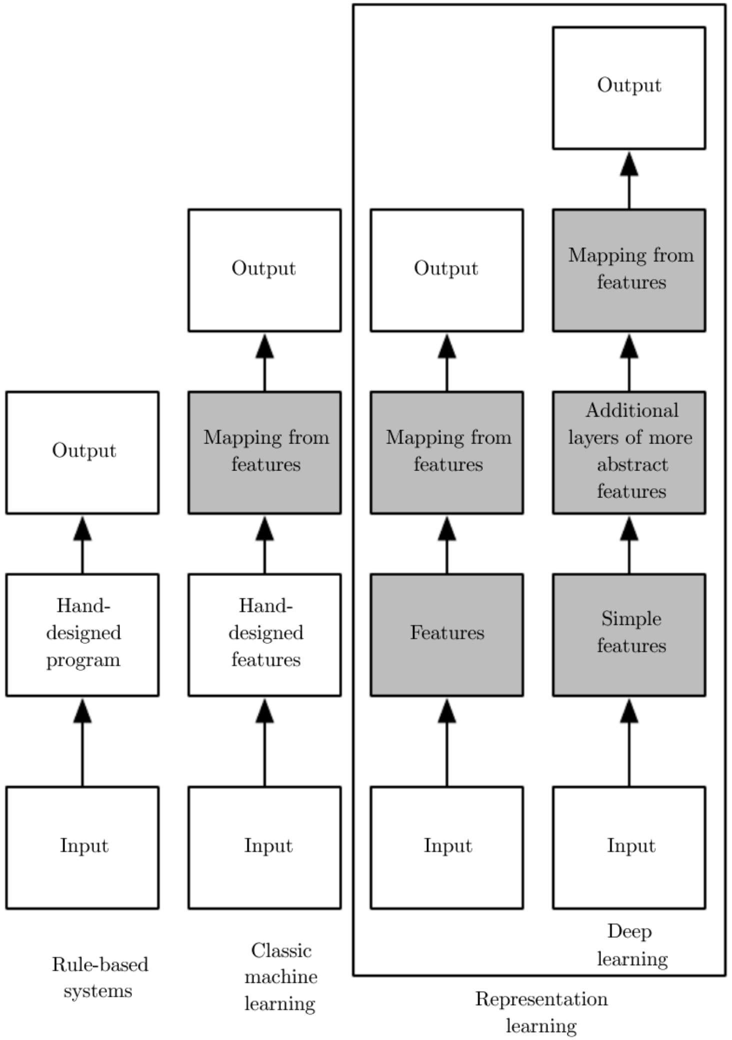
Improving performance of a system on future tasks based on observations/data about the population

1. Extract a set of features from each population instance.
2. Learn relationships between features and labels.

TRADITIONAL MACHINE LEARNING

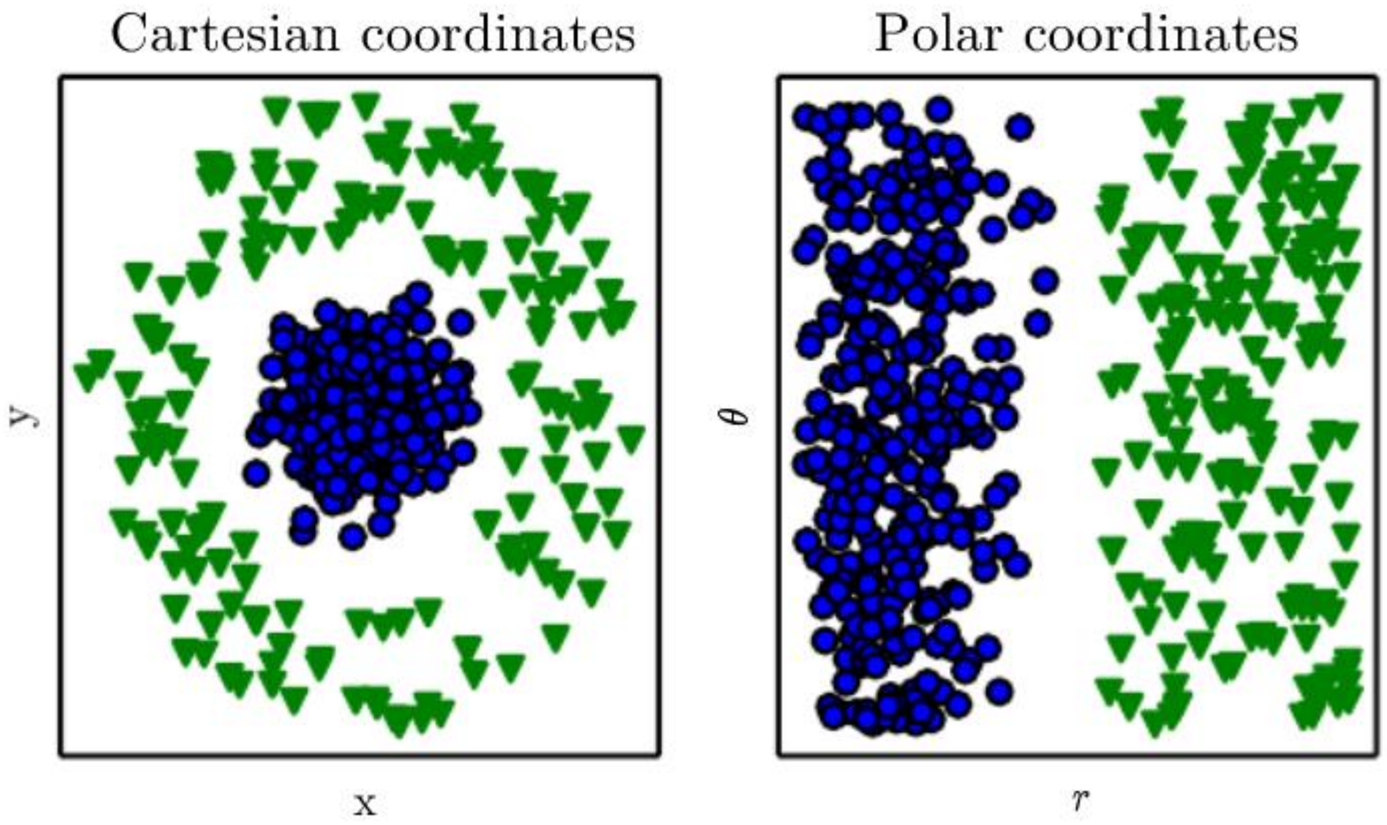
- Depends on the features we extract (how we represent inputs)

REPRESENTATION LEARNING



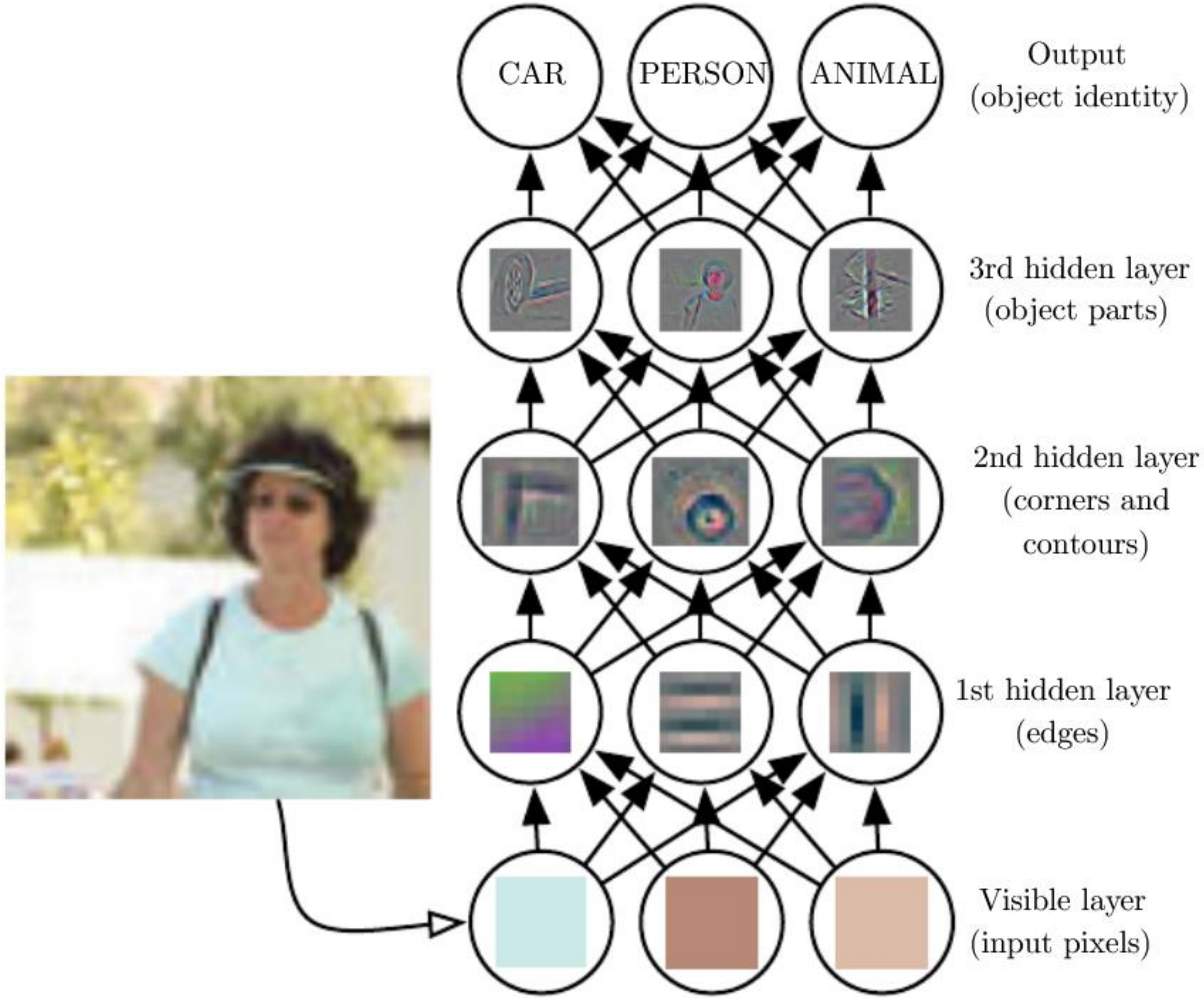
Goodfellow et al., Deep Learning Book, 2016.

POWER OF REPRESENTATION



Goodfellow et al., Deep Learning Book, 2016.

DEEP LEARNING



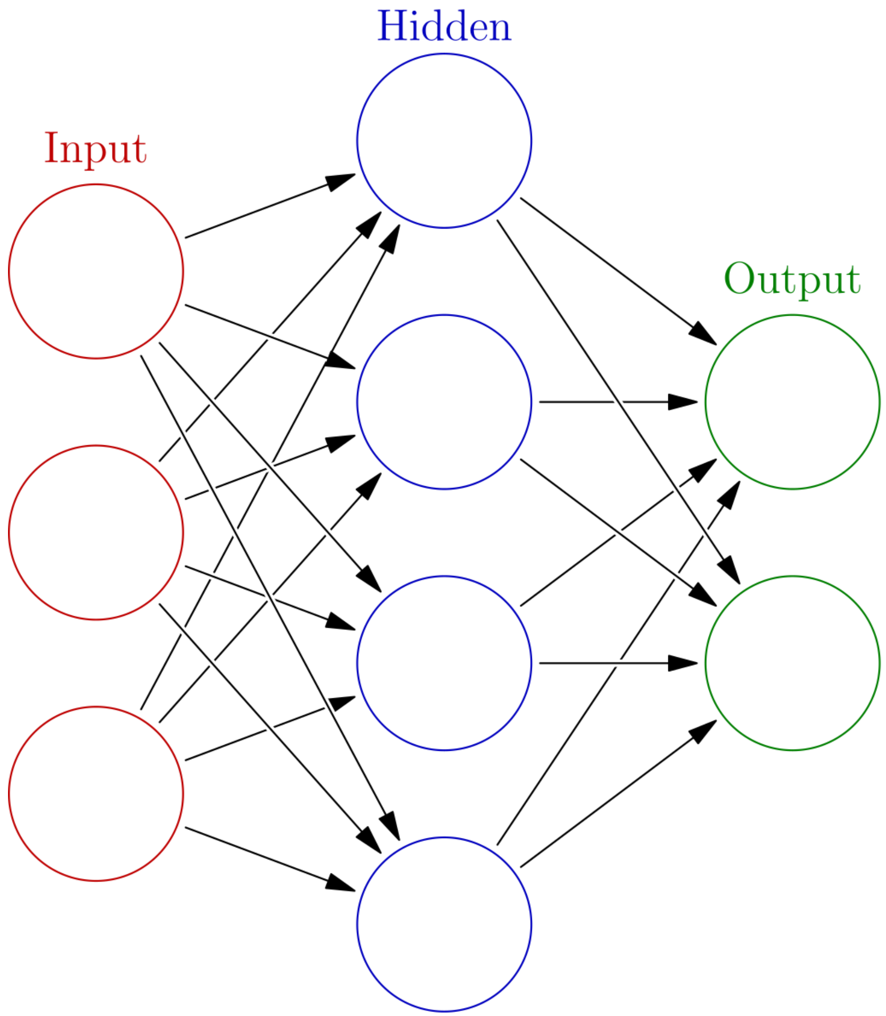
Goodfellow et al., Deep Learning Book, 2016.

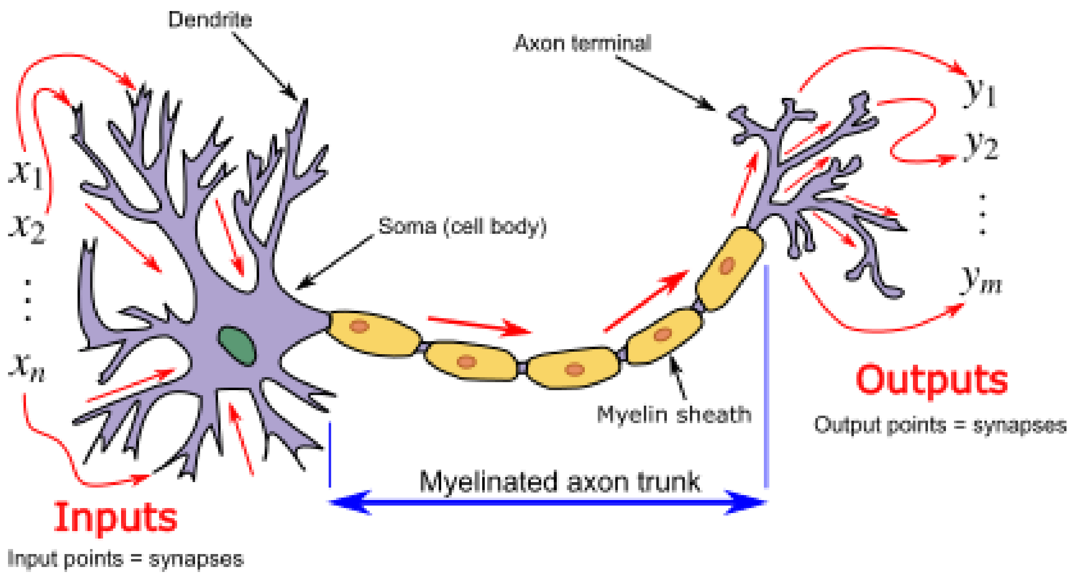
WHAT'S SO DEEP ABOUT DEEP LEARNING?

- Repeated composition of functions

- Learn complex representations out of simple representations

DEEP LEARNING VS. NEURAL NETWORKS





Images: https://en.wikipedia.org/wiki/Artificial_neural_network